Contract-Carrying Analytical Physics Receipts for Bounded CAD Screening: A Controlled Conformance Study

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October 2026

Abstract

Analytical engineering calculations are often copied away from the geometry, material declarations, and load assumptions on which they depend. This paper specifies and evaluates a small, fail-closed receipt that binds a versioned analytical result to canonical CAD-primitive identity, an explicit material card, and a named load case. The mechanism performs exact semantic checks only where the mapping is unambiguous and can later test whether the selected geometry has changed. We compare a full binding with a hash-only ablation and an unbound analytical request in a preregistered synthetic conformance matrix. The locked 216-case run was classified `SUPPORTED_WITHIN_CONTROLLED_CONTRACT`; the predeclared arm-specific endpoints recorded 528/528 expected outcomes. For post-result mutation, the full-versus-unbound discordance was 48 to 0 (exact two-sided McNemar $p = 7.11e-15$). The study concerns software-contract conformance, not physical accuracy, solver validity, safety, or enterprise CAD/CAE generalization.

1 Introduction

Digital-thread and simulation-provenance research has long argued that model artifacts, assumptions, executions, and results should remain traceable across the engineering lifecycle [1, 2, 3, 4]. Nevertheless, a lightweight analytical calculation can still be preserved as an isolated JSON or spreadsheet result with no executable link to the exact geometry, material declaration, or load case that produced it. The resulting number may remain syntactically valid after its dependencies change.

This work asks a deliberately narrow question: for closed-form calculations over a bounded set of CAD primitives, can a machine-readable receipt detect predeclared contradictions and post-result geometry drift while retaining deterministic valid-case behavior? We contribute (1) a strict open reference contract; (2) an implementation in NeuroCAD; and (3) a preregistered controlled ablation across full, hash-only, and unbound arms. We do not claim to invent provenance, hashing, CAD/CAE associativity, or the digital thread. We also do not claim that correct provenance makes an analytical model physically valid.

2 Related work and novelty boundary

Hedberg et al. frame the digital thread around lifecycle traceability and reuse of authoritative product data [1]. NIST reference architectures address contextual integration among heterogeneous engineering systems [2, 3], while the W3C PROV data model supplies a general representation of entities, activities, agents, and derivations [4]. Simulation-specific work captures provenance across tools and workflow stages [5, 6, 7], and NIST has studied traceability between systems-engineering and finite-element artifacts [8]. RO-Crate provides a practical packaging model for research objects and their metadata [9].

Accordingly, the contribution here is not an algorithmic novelty claim. It is an executable, bounded reference contract and a controlled conformance study of one implementation choice: bind a small analytical

result to canonical primitive geometry plus explicit material and load declarations, then fail closed on exact contradictions or later geometry drift.

3 Receipt contract

Let an analytical request be $R = (m, x, c)$, where m is a versioned analytical model, x is a strict mapping of numeric inputs whose names encode units, and c optionally contains geometry, material, and load bindings. Canonical JSON uses lexicographically sorted object keys, compact separators, UTF-8 encoding, and forbids non-finite numbers. For any value z,

$$H(z) = \text{SHA256}(\text{canonical_json}(z)). \tag{1}$$

The returned receipt records the executed model version, exact inputs and outputs, disclosed assumptions, claim boundary, $H(R)$, $H(x)$, and hashes of each declared context object. A hash establishes content identity only; it is not a signature, an authority proof, or evidence that the content is correct.

3.1 Geometry identity

For one selected component, the geometry-relevant tuple is

$$G = (\text{id}, \text{part}, \text{operation}, \text{shape}, \text{parameters}, \text{translation}, \text{rotation}). \tag{2}$$

Display labels and project revision counters are excluded. Re-verification reloads the component and returns *current* exactly when its observed $H(G)$ equals the receipt. The implementation reports closed-form isolated- primitive volume, surface area, centroid, transformed axis-aligned bounds, and selected unambiguous section properties. These are not Boolean part or assembly properties. Mass is reported only for an additive primitive with an explicit density.

3.2 Material, load, and exact semantic checks

A versioned material card declares a source and finite properties; a named load case declares a coordinate system, source, and values. The evaluator does not select materials, interpolate curves, infer directions, combine loads, or invent safety factors. Repeated material or load values must numerically agree with model inputs to relative tolerance 10^{-12}.

Geometry/model checks are restricted to three exact mappings: cylinder diameter and length for circular-shaft torsion, isolated displaced volume for hydrostatic buoyancy, and isolated volume and total surface area for a lumped thermal transient. Ambiguous quantities such as effective buckling length, convection face area, and internal pipe diameter are never inferred.

4 Experimental design

The protocol was written before protected outcome access. The independent unit is one generated case. Repeated evaluation of a case is a determinism check, not an additional independent unit. The fixed seed is 20261011.

All applicable deterministic contract endpoints require 100% conformance; a single miss falsifies the corresponding claim. We report exact numerators and denominators. Wilson 95% intervals describe the fixed matrix and are not used as population inference. A paired exact two-sided McNemar test summarizes full-versus-unbound post-mutation discordance, but the decision rule remains the all-cases threshold. Timing, resident memory, and report size are host-only diagnostics, not cross-system performance benchmarks.

The protected run requires a clean frozen Git revision, exact source and case hashes, at least 512 MiB of free plus speculative memory, at least 2 GiB of output storage, one process, a new output directory, and an absent execution ledger. The ledger is written before case execution so a crash consumes the single attempt. Runtime errors and mismatches remain outcomes; no failed case may be removed.

Table 1: Preregistered confirmatory matrix.

Case family	Count
Valid bound requests	48
Geometry/input contradictions	24
Material/input contradictions	24
Load/input contradictions	24
Stale expected geometry hashes	24
Post-result geometry mutations	48
Non-geometric mutations	24
Total	216

Table 2: Mechanism ablation. A dash means that the arm has no such mechanism.

Capability	Full	Hash-only	Unbound
Canonical component identity	yes	yes	–
Semantic geometry checks	yes	–	–
Material/load agreement checks	yes	–	–
Post-result freshness API	yes	yes	–

5 Results

The single protected execution completed 216 cases and was classified `SUPPORTED_WITHIN_CONTROLLED_CONTRACT`. Table 3 reports every predeclared arm-specific endpoint. The intervals are descriptive Wilson 95% intervals for the fixed matrix.

For the paired post-result geometry mutations, the full arm alone detected 48 cases and the unbound arm alone detected 0 cases; the exact two-sided McNemar value was $p = 7.10543e-15$. This comparison describes the presence or absence of the preregistered dependency mechanism and is not a population-effect estimate.

On this host, median recorded successful evaluation times were 0.619 ms for full binding ($n = 120$), 0.248 ms for hash-only ($n = 192$), and 0.013 ms for unbound requests ($n = 216$). Failed evaluations have no elapsed-time record, so these samples are conditional and must not be compared as balanced performance arms. Corresponding p95 times were 1.146, 0.534, and 0.034 ms. The median serialized full receipt was 3902 bytes and the maximum was 4199 bytes. These are local diagnostics only.

Table 3: Confirmatory contract endpoints. "Expected outcome" includes both detection where a mechanism applies and declared non-detection where an arm has no such mechanism.

Endpoint	Expected outcomes	Wilson 95% interval
Valid full acceptance	48/48	0.926–1.000
Valid full determinism	48/48	0.926–1.000
Full geometry contradiction detection	24/24	0.862–1.000
Full material contradiction detection	24/24	0.862–1.000
Full load contradiction detection	24/24	0.862–1.000
Full stale-hash detection	24/24	0.862–1.000
Hash-only semantic non-detection	72/72	0.949–1.000
Unbound semantic non-detection	72/72	0.949–1.000
Full post-mutation detection	48/48	0.926–1.000
Hash-only post-mutation detection	48/48	0.926–1.000
Unbound post-mutation non-detection	48/48	0.926–1.000
Full non-geometric specificity	24/24	0.862–1.000
Hash-only non-geometric specificity	24/24	0.862–1.000

6 Threats to validity

Construct validity. Several endpoints are intentionally contractual: a content hash should change when hashed geometry changes. The study tests implementation conformance, not a new physical phenomenon. An unbound arm's failure to detect drift follows from its lack of a geometry dependency and should not be interpreted as a defect in all unbound calculation workflows.

Internal validity. The implementer authored the code, generator, and cases. A frozen manifest, single-attempt ledger, retained raw records, exact hashes, and fail-closed gates limit post-outcome flexibility, but do not supply independent authorship or adjudication.

External validity. The cases use synthetic bounded primitives and three exact geometry/model mappings. They do not represent full CSG, assemblies, meshing, nonlinear solvers, enterprise PLM, or heterogeneous commercial CAD/CAE environments. No physical experiment or qualified solver comparison is included.

Conclusion validity. Wilson intervals are descriptive for this fixed matrix. Timing on one host and small JSON artifacts cannot support a general performance claim. Any confirmatory failure remains negative or mixed evidence under the frozen rule.

7 Discussion

The practical value of the contract is modest but concrete: it makes a result's declared dependencies inspectable and gives downstream automation a fail-closed freshness test. Its limitations are equally important. Provenance cannot repair an inappropriate analytical model, an unreliable source, an omitted load, or an invalid safety argument. For higher-consequence use, the receipt would need qualified source governance, solver verification and validation, uncertainty analysis, access control, signatures, and domain review.

8 Reproducibility and artifact availability

The implementation, protocol, generator, raw JSONL records, metrics, manifest, execution ledger, and paper-generation script are retained in the NeuroCAD repository. The result-bearing source revision is `786ba9ba5c2249a710529c3b07bf82d4e29c7f36`. The artifact records Python and platform identity, exact

file hashes, case-manifest hashes, raw-result and metric hashes, resource admission, and Git state. A clean-environment reproduction can check software behavior, but is not labeled independent external validation unless performed and adjudicated by an independent party.

9 Conclusion

This study narrows a broad digital-thread ambition to a testable local contract: an analytical result names and hashes the primitive geometry, material declaration, and load case it depends on, checks exact agreements at evaluation time, and exposes later geometry drift. The confirmatory conclusion is limited to the preregistered synthetic matrix and the stated implementation boundary. It does not establish physical correctness or engineering safety.

References

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